

Lecture 2A - Causal Graph

A New Language of Cause and Effect

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- 1 The graph terminology
- 2 Causal graph
- 3 The do-operator
- 4 Understanding do-calculus

Causation $\neq$ correlation

- Try to explain how causation and correlation are different.

Causation $\neq$ correlation

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- You know the differences but it is difficult to verbalize.

Causation $\neq$ correlation

- Try to explain how causation and correlation are different.
- You know the differences but it is difficult to verbalize.
- We lack a sort of “language” to explain the differences.

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From a causal claim to a graph

Causal question

Does **AI adoption** cause researchers to concentrate science on a narrower set of topics?

Core variables

- **AI adoption (X):**
researcher uses AI tools.
- **Scientific focus (Y):**
breadth vs. concentration of topics.

Supporting constructs

- **Resources & data availability** (Z): labs, ability, networks, and data-rich fields.
- **Topic choice** (M): movement toward data-rich or established topics.

Alternative explanations under the observations

Observation

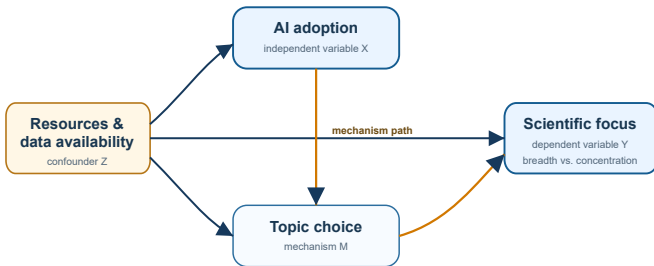
AI adoption is positively correlated with the concentration of topics.

Alternative explanations

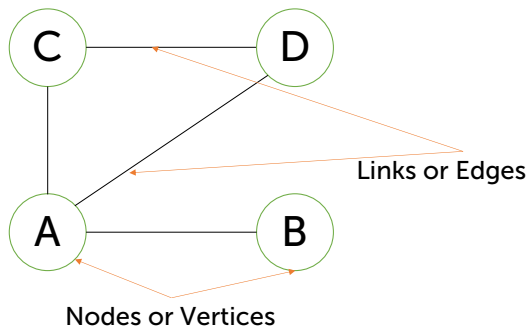
- AI adoption $\rightarrow$ scientific focus
- AI adoption $\leftarrow$ resources & data availability $\rightarrow$ scientific focus
- AI adoption $\rightarrow$ topic choice $\rightarrow$ scientific focus

From theory to graph

The graph turns concepts into variables, turns theory into arrows, and **reveals the empirical challenges**.

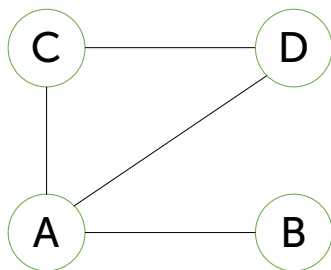


The atoms of a graph

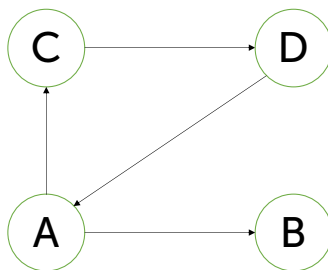


- **Nodes:** any entity, e.g., people, organizations, charging stations, internet routers
- **Links:** any relationship or connectedness

Directed vs. undirected graphs



An undirected graph



A directed graph

- Undirected graph: no directions, e.g., Facebook friends
- Directed graph: with directions, e.g., Twitter followers

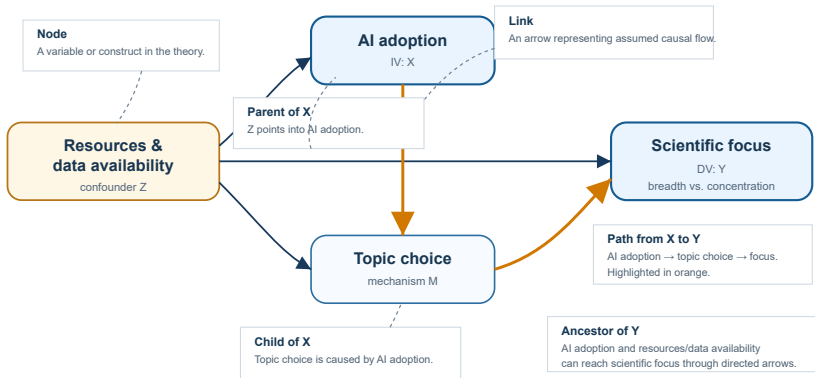
Paths in directed graphs

- Between two nodes X and Y in a directed graph:
 - Adjacent: $X \rightarrow Y$ or $Y \rightarrow X$
 - Non-adjacent: X Y
- If $X \rightarrow Y$, X is a parent of Y and Y is a child of X .
 - If $X \rightarrow \dots \rightarrow Y$, X is an ancestor of Y , and Y is a descendant of X .

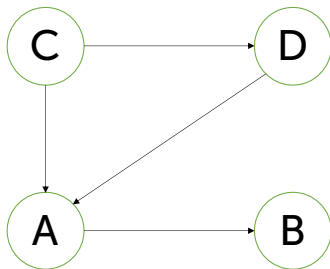
Definition (Paths in directed graphs)

A path in a directed graph is any sequence of adjacent nodes, regardless of the direction of the links that join them.

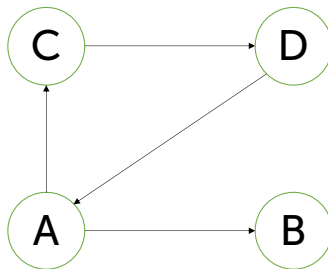
Graph terms in the example DAG



Cycles in directed graphs



A directed acyclic graph (DAG)



A directed cyclic graph

Definition (Cycles in directed graphs)

A cycle is a particular type of path that starts from a node and links back to the same node, i.e., $X \rightarrow \dots \rightarrow X$.

1 The graph terminology

2 Causal graph

From conceptual model to causal graph

Basic structures of causal graph

③ The do-operator

4 Understanding do-calculus

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From conceptual model to causal graph

Basic structures of causal graph

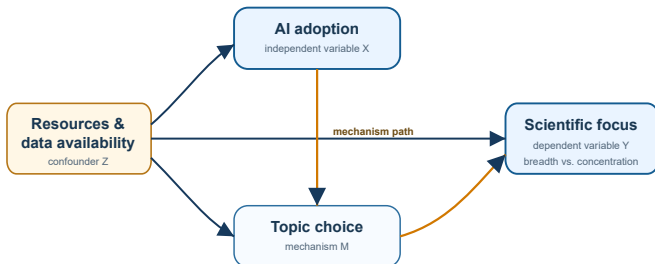
③ The do-operator

4 Understanding do-calculus

From conceptual model to causal graph

A DAG lets us link our theory to empirical questions:

- What is the causal effect we want to estimate?
- What are the alternative explanations?
- Which paths create confounding?
- Which variables should we control for?
- Which controls would create bias?



What is a causal graph?

Simply put it: a causal graph is a directed acyclic graph where

- The nodes represent variables.
- The links represent causal flow.

Definition (Causal Links)

A variable X is said to be the cause of a variable Y , if an **exogenous** change in X can **flow** on the graph to Y .

What assumptions make a DAG useful?

For a DAG to guide causal inference, we need three assumptions:

- 1 **Arrows mean direct causes.** If $X \rightarrow Y$, then X is a direct cause of Y in our theory.
- 2 **Parents screen off other non-effects (a.k.a. local Markov assumption).** Once we know the direct causes of a variable, other variables should not provide extra information about it.
- 3 **No accidental cancellations.** The statistical patterns we observe should reflect the causal structure, not the exact offsetting effects.

In short

The graph is not just a picture. It bridges causation and correlation.

Why the local Markov assumption matters?

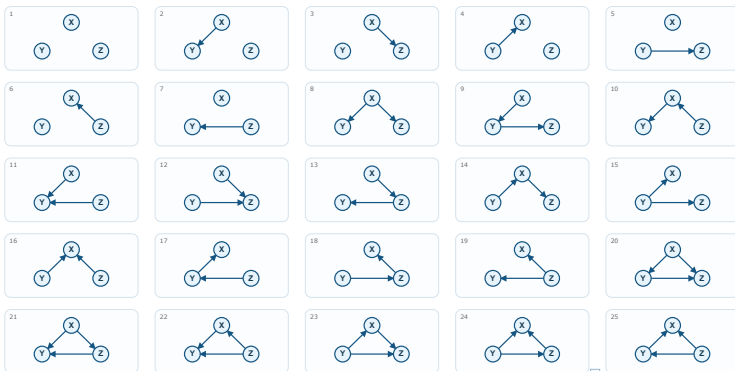
Real causal systems can be very complex.

2 variables: 3 possible DAGs



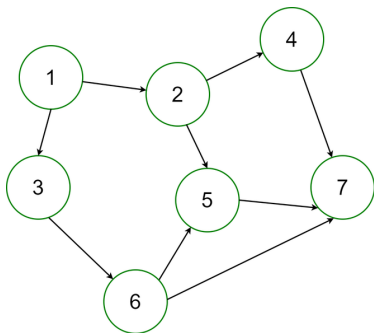
3 variables: 25 possible DAGs

For each pair, there can be no edge, an arrow one way, or an arrow the other way. Cycles are ruled out.

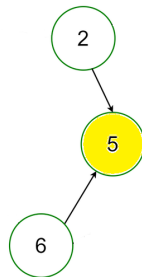


Why the local Markov assumption matters?

The local Markov assumption allows us to focus on two layers - a node and its parents.



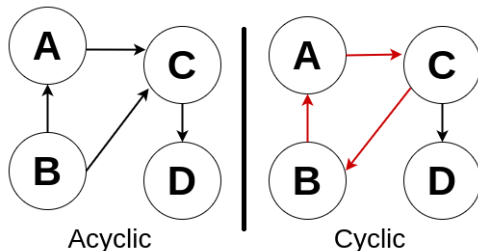
(a) Full DAG



(b) DAG of Node 5

The limitation of DAG

DAG assumes no cycles exist.



Business interpretation

Many business systems have feedback loops in reality. For causal identification, we usually study one direction at one time, over a specified time window or the *partial equilibrium*.

1 The graph terminology

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From conceptual model to causal graph

Basic structures of causal graph

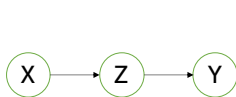
3 The do-operator

4 Understanding do-calculus

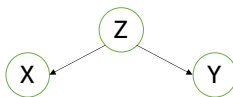
Starting with motifs: 3 motifs

The local Markov assumption allows us to reason locally: **use small structures to understand larger DAGs.**

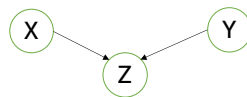
- We focus on motifs - repeated structures on graphs.
- Three-node motifs: chains, forks, and collisions (or immorality)



(a) Chain



(b) Fork

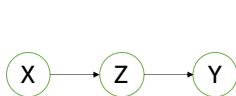


(c) Collision

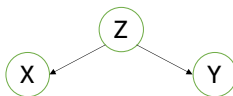
Starting with motifs: some observations

...between X and Y	Chains	Forks	Collisions
Association	> 0	> 0	$= 0$
Causation	> 0	$= 0$	$= 0$

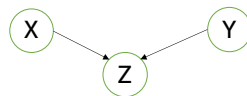
Table 1: Association vs. causation in 3 motifs



(a) Chain



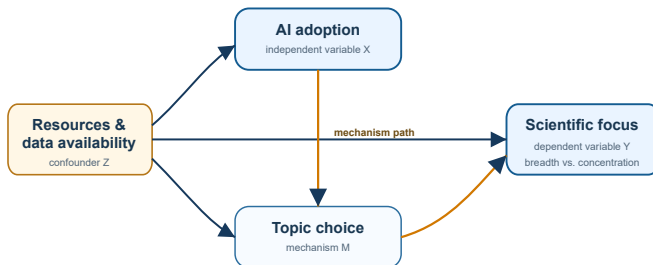
(b) Fork



(c) Collision

Aligning caution with correlation

The observed association may come from different paths (alternative explanations) in the DAG:



The problem

We do not know which path generates the observed association. To estimate a causal effect, we need to isolate the causal path and **block unwanted paths**.

Starting with motifs: blocking paths

- For a fork $X \leftarrow Z \rightarrow Y$, X and Y are correlated but $X \nrightarrow Y$.
- But how do we know this if we only observe data or $\gamma(X, Y)$?
- Solution: **What if we remove the arrow into X ($Z \rightarrow X$)?**

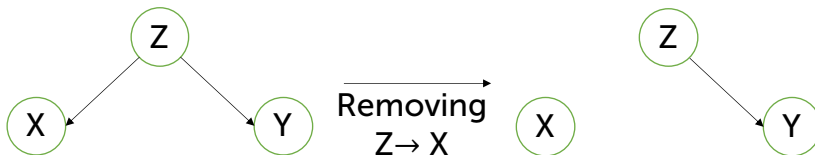


Figure 1: Blocking incoming paths to X

Starting with motifs: blocking paths

If we want to know the effect of X on Y , we must remove all “redundant paths.”

- In formal terms, we “block” all unwanted paths.

How to block paths between X and Y ?

- Let’s look at the motifs one by one.

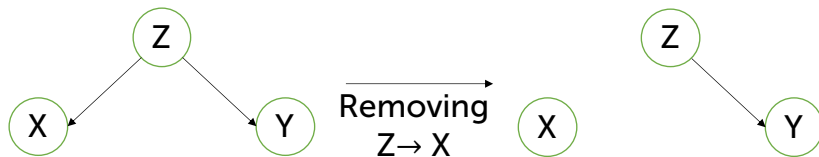
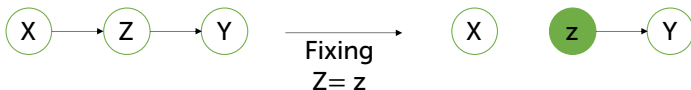


Figure 2: Blocking incoming paths to X

Blocking paths: chains

Chains (front-door paths)

To block a chain $X \rightarrow Z \rightarrow Y$, we fix the value of Z (a.k.a. the mediator) or set $Z = z$.



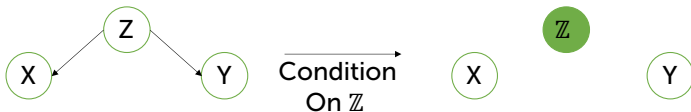
Testing with causal links assumption

A change in ΔX cannot pass through to Y .

Blocking paths: forks

Forks (back-door paths)

To block a fork $X \leftarrow Z \rightarrow Y$, we condition on the full support of $Z \in \mathbb{Z}$ (a.k.a. the moderator).



Testing with causal links assumption

Given Z , the value of X and Y are pre-determined.

A change in ΔX therefore does not pass over to Y .

Blocking paths: collisions

Collisions (immoral paths)

A collision path $X \rightarrow Z \leftarrow Y$ is already blocked.



Testing with causal links assumption

A change in ΔX changes Z , but NOT Y .

Blocking paths: more on collisions

Do not condition on collider Z

It is tempting, but conditioning on Z creates spurious correlation between X and Y !



Sample selection view

- In the general population, gender and age are independent.
- In the sub-population of CEOs, more older males.
- In the sub-population of nurses, more younger females.

Motifs as control-variable rules

Motif	Role of Z	If we control for Z	Empirical lesson
$X \rightarrow Z \rightarrow Y$	Mechanism / mediator	Blocks the mechanism	Do not control for Z if you want the total effect
$X \leftarrow Z \rightarrow Y$	Confounder / common cause	Blocks back-door path	Control for Z
$X \rightarrow Z \leftarrow Y$	Collider / selection variable	Opens a noncausal path	Do not control for Z

The same regression control can be helpful, harmful, or conceptually inappropriate depending on its position in the DAG.

D-separation: the general rules of path-blocking

Definition (Blocked Paths)

A path between nodes X and Y is blocked by a node W along the path if either of the following is true:

- 1 *For a chain along the path, $\cdots \rightarrow W \rightarrow \cdots$, W is fixed.*
- 2 *For a fork, $\cdots \leftarrow W \rightarrow \cdots$, W is conditioned on.*
- 3 *For a collision $\cdots \rightarrow W \leftarrow \cdots$, W is not conditioned on, as well as all descents of W .*

D-separation

Two nodes X and Y are d-separated by a set of nodes Z , if all paths between X and Y are blocked by Z .

D-separation as an empirical design tool

When estimating the causal effect of X on Y , use the DAG to audit the control strategy.

- ① List all paths between X and Y .
- ② Apply the d-separation rules:
 - Keep the causal paths open if estimating the total effect.
 - Block back-door paths through confounders.
 - Do not open collider paths.
- ③ Conclude controls are confounders, mechanisms, or selection variables.

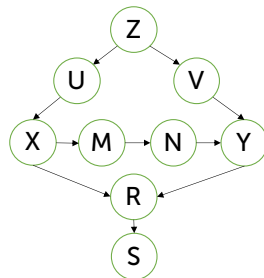
D-separation: an application

First, we find all the three types of paths:

1. Backdoor paths: $X \leftarrow U \leftarrow Z \rightarrow V \rightarrow Y$.
2. Frontdoor paths: $X \rightarrow M \rightarrow N \rightarrow Y$.
3. Collision paths: $X \rightarrow R \text{ (de}(R)) \leftarrow Y$.

Second, we construct a blocking set that:

1. Includes one or two or three of $\{U, V, Z\}$.
2. Includes one or two of $\{M, N\}$
3. Does NOT include R and S (descendant of R)



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2 Causal graph

3 The do-operator

Interventions: the secret sauce

The $\text{do}(\cdot)$ operation

4 Understanding do-calculus

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Interventions / Manipulations

When they are infeasible...

- Randomized control experiments - **the golden standard**.
- However, not everything can be intervened (ethically).
 - Example: age, gender, or some medical treatments etc.
- In business or economics studies:
 - Something is simply too costly to intervene.
 - e.g., lowering price of electricity? Tax cuts?

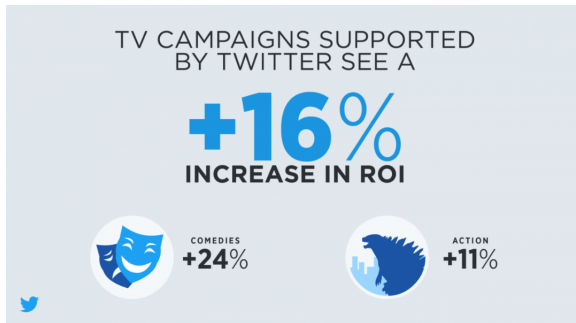
What then?

- People turn to observational data.
 - “Conditional on...” or “adding control variables...”
- Difficult to untangle causation from correlation.
- What's worse:
 - We see patterns and are drawn towards causality.
 - **It is imperative to evaluate causal claims, but how?**

Twitter example

New movie marketing research reveals Twitter Ads deliver increased ticket sales (LINK)

- “The research team used multivariate regression analysis (a process that measures and predicts the sales impact of various media channels) to understand the effects of changes in Twitter media for movies.”



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Interventions: the secret sauce

The $\text{do}(\cdot)$ operation

4 Understanding do-calculus

The do-operator

- Do-operator formalizes manipulations in causal graphs.
 - Manipulating X : $\text{do}(X) = x$ or setting $X = x$.
- It's the first step to evaluate causal claims.
- Graphical interpretation: **to remove all arrows into X .**

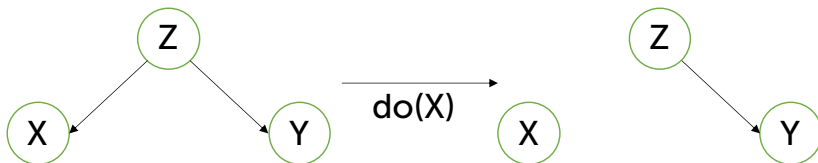


Figure 3: Do-operation $\iff$ Manipulation

$P(Y | X = x)$ vs. $P(Y | \text{do}(X) = x)$

Comparing $P(Y | X = x)$ vs. $P(Y | \text{do}(X) = x)$

$P(Y X = x)$	$P(Y \text{do}(X) = x)$
Conditioning	Intervening
Purely Statistical	Beyond statistics
Select a subgroup	Create a population
Always calculable	Not always calculable
In the full DAG	In a surgically-altered DAG (subgraph)

Why is $P(Y | X)$ not always $P(Y | \text{do}(X))$?

Researcher	X: AI adoption	Y: breadth of focus
1	1	2.1
2	1	2.4
3	1	2.6
4	1	2.9
5	1	3.0
6	0	3.5
7	0	3.8
8	0	4.1
9	0	4.2
10	0	4.4

From the table, $E[Y | X = 1] = 2.6$ and $E[Y | X = 0] = 4.0$.

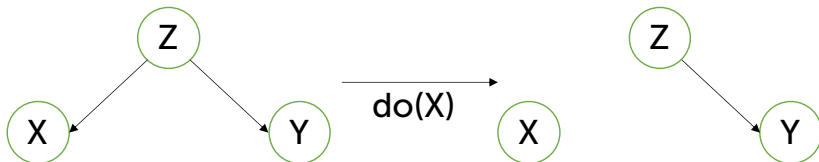
But what we learned?

From the data, we know $P(Y | X)$, but not $P(Y | \text{do}(X))$, unless AI adoption is manipulated.

How to obtain $P(Y \mid \text{do}(X) = x)$?

Procedure to obtain $P(Y \mid \text{do}(X) = x)$

- 1 Do a **surgery on the full DAG** by removing all arrows into X
- 2 Transform the conditional probability with **the new DAG**
 - 1 i.e., $P(Y \mid \text{do}(X) = x) = P(Y \mid X)$
- 3 Calculate the conditional probability $P(Y \mid X) = P(Y)$



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The definition of do-calculus

Definition (Do-calculus)

The do-calculus is an axiomatic system for replacing probability formulas containing the do operator with ordinary conditional probabilities.

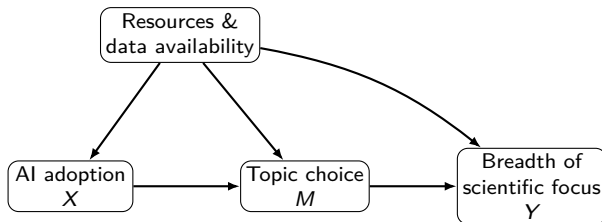
- The normal reaction to the definition is “**say what?**”
- Let’s look at the intuitions behind it!

The intuitions behind do-calculus

- To know causal effects, we need $P(Y \mid \text{do}(X))$.
- E.g., Average Treatment Effect:
 $P(Y \mid \text{do}(X) = 1) - P(Y \mid \text{do}(X) = 0)$.
- We need to manipulate X to get $P(Y \mid \text{do}(X))$, but in many cases, we cannot.
- We can have $P(Y \mid X)$, but $P(Y \mid X) \neq P(Y \mid \text{do}(X))$.

Illustration: AI adoption on research focus

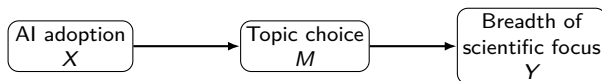
Let X be AI adoption and Y be breadth of scientific focus.



- ...But we cannot force researchers to adopt AI.
- The DAG says Z (resources and data availability) create a back-door path.

Illustration: AI adoption on research focus

Let X be AI adoption and Y be breadth of scientific focus.



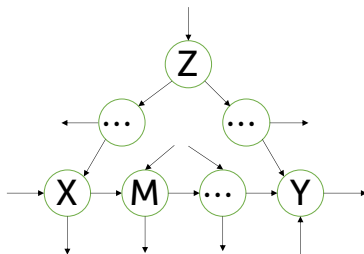
- If we condition on Z , we block the non-causal paths or surgically remove Z .
- In this new DAG, $P(Y | X) = P(Y | \text{do}(X))$.
- So, observing and conditioning on Z is the essential condition.

Generalizing the intuition

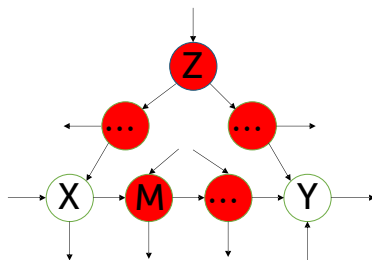
In practice, the DAG may be **much more complex**.

Do-calculus: which $\text{do}(\cdot)$'s or observations for other variables are essential if

$$P(Y \mid \text{do}(X)) = P(Y \mid X)?$$



(a) A complex DAG



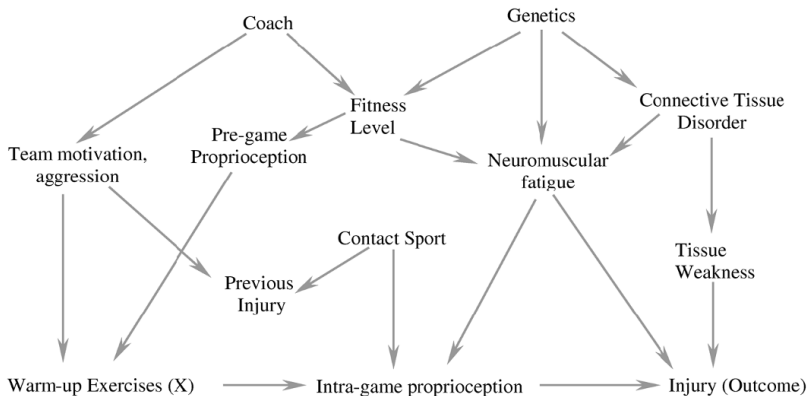
(b) Which one(s) to manipulate?

Three rules of do-calculus

- Omitted here. Please see **Pearl 1995** for more details.
- Proved by **Shpitser and Pearl 2006**, the rules are complete.
 - That is, the three rules are sufficient to transform $P(Y \mid \text{do}(X), \dots) \mapsto P(Y \mid X, \dots)$.
 - If the rules are not met, $P(Y \mid \text{do}(X), \dots)$ may not be known from the DAG.
- For us: **causal identification for business research**.
 - The implications from the three rules of do-calculus.

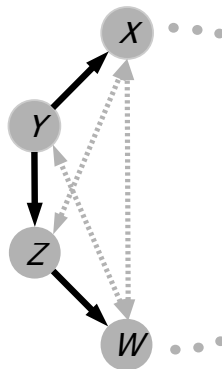
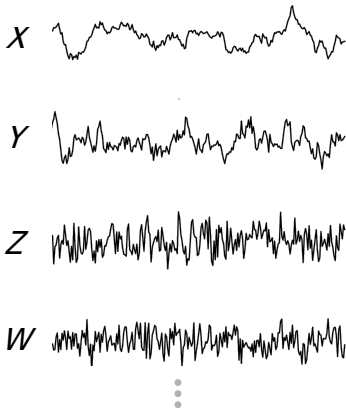
To sum up: Where do DAGs come from?

- From human minds or “theories”.



To sum up: Where do DAGs come from?

- **From data with causal discoveries** (using do-calculus).



Extra Slides

Technical Details

The “faithfulness” assumption

- The causal link assumption is not sufficient to have a unique DAG. We need the faithfulness assumption to deal with tricky scenarios.

An example

- Consider a DAG, with $X \rightarrow Y \leftarrow Z$, and we further assume $Y = X + Z$, without the loss of generality.
- Conditional on $Y = y$, the covariance between X and Z should be **non-zero**.
 - $\text{cov}(X, Z \mid Y = y) = \text{cov}(X, y - X) = yE(X) - E(X^2)$.
- If $yE(X) - E(X^2)$ is always 0, then $\text{cov}(X, Z \mid Y = y) = 0$.
- If $\text{cov}(X, Z \mid Y = y)$ is always 0, then the DAG $X \rightarrow Y \leftarrow Z$ cannot be right. But it is the true DAG.
- We therefore have the assumption to rule the situation.

Do-operator: some formality

Definition (Do-operator)

If we intervene on a set of nodes S on a DAG, then for all nodes X_i , we have the following:

- ① For $X_i \notin S$, $P(X_i \mid pa(X_i))$ remains unchanged.
- ② For $X_i \in S$, if $x_i = do(X_i)$, then $P(X_i = x_i \mid pa(X_i)) = 1$, and otherwise $P(X_i = x_i \mid pa(X_i)) = 0$.

Do-operator: some formality

Corollary (Truncated Factorization)

We assume that P and G satisfy the Markov and causal links assumption. Given a set of intervened nodes S , we have,

$$P(X_1, X_2, \dots, X_n \mid do(S = s)) = \prod_{X_i \notin S} P(X_i \mid pa(X_i))$$

The three rules of do-calculus

For a DAG G , let $G(\neg\rightarrow_X)$ denote the graph G with all incoming links to X removed. Similarly, $G(\neg\leftarrow_X)$ denote the graph G with all outgoing links from X removed. Combining these two, the graph G with incoming links to X and outgoing links from Z removed is denoted as $G(\neg\rightarrow_X\neg\leftarrow_Z)$. In addition, I will use $\perp_G$ to represent d-separation on the graph.

Theorem (The Rules of Do-calculus)

Given a DAG G , an associated distribution P , and disjoint set of variables Y , X , Z and W , the following rules hold:

- ① $P(Y \mid do(X), Z, W) = P(Y \mid do(X), W)$, if $Y \perp_{G(\neg\rightarrow_X)} Z \mid X, W$.
- ② $P(Y \mid do(X), do(Z), W) = P(Y \mid do(X), Z, W)$, if $Y \perp_{G(\neg\rightarrow_X\neg\leftarrow_Z)} Z \mid X, W$.
- ③ $P(Y \mid do(X), do(Z), W) = P(Y \mid do(X), W)$, if $Y \perp_{G(\neg\rightarrow_X\neg\rightarrow_{Z \setminus \text{Ancestor}(W)}} Z \mid X, W$, where $Z \setminus \text{Ancestor}(W)$ denotes non-ancestors of W in set Z .

The intuition for the 3rd rule

Again, we remove the $do(X)$, and get

$$P(Y \mid do(Z), W) = P(Y \mid W), \text{ if } Y \perp_{G(\neg\rightarrow_{Z \setminus \text{Ancestor}(W)})} Z \mid W$$

This implies if we remove the do-operation $do(Z)$ by reintroducing the links that come into Z , we would not cause flow of association into Y . Because $do(Z)$ removes all incoming links to Z give us $G(\neg\rightarrow_Z)$, the main pathways for the flow of association between Z and Y are the frontdoor paths from Z to Y in $G(\neg\rightarrow_Z)$. If W contain all chain nodes, naturally Y and Z will be d-separated. However, if W also contains colliders (i.e., nodes that are descendants of Z), conditioning would lead to non-causal association flow between Y and Z . Therefore, we have to modify the set of Z and limit the set of manipulated nodes in Z to those that are not ancestors of W . By doing so, W would not contain colliders or descendants of colliders. Simply put, Rule 3 applies the do-separation criterion on colliders.

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